Autonomous Networking

Networks that observe, decide, and adapt

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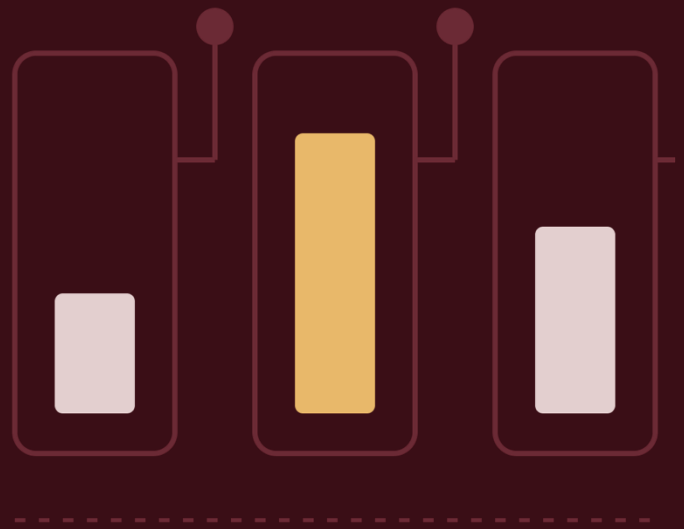
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AUTONOMOUS NETWORKING · A.Y. 26-27

Network technologies · drones

Networks of drones



UAVs, MAC and routing in flying ad hoc networks, and periodic data offloading with Stop & Route

WHERE WE ARE

Three threads, one goal

Networks	RL basics	RL-based protocols
RFID ✓	RL and bandits ✓	MAC
Sensor networks ✓	Testbed and UCB ✓	Routing
Drones · today	MDPs · next	Project
IoT, robophysical	Q-learning	

Goal of the course: protocols that adapt by themselves, for networks that change by themselves. Drones are the most dynamic network we will meet.

TODAY'S QUESTION

How do we move data across a network whose nodes fly?

USE THE NETWORK

Radios, MAC and **routing** when links appear and disappear in seconds

MOVE THE NODES

Drones can fly where they are needed: **move them to make the network connected** (Stop & Route)

TODAY'S PLAN

Drones, communication, moving to connect

Part 1 · Drones

- → **Drones and applications**
- ○ Dronets: four problems

Part 2 · Communication

- ○ Radios and MAC
- ○ Routing: families, B.A.T.M.A.N.
- ○ Geographic routing

Part 3 · Moving to connect

- ○ Stop & Route
- ○ From protocols to decisions

Unmanned aerial vehicles



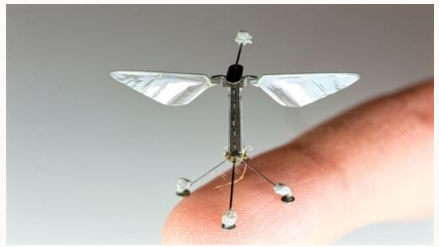
- A UAV, or **drone**, is an aircraft **without a human pilot on board**
- It may fly with various degrees of autonomy:
  - **remote control** by a human operator
  - **autonomous flight** with on-board computers

In this course

Autonomous drones: they also take their networking decisions by themselves

A very heterogeneous family

feature	range
weight	0.5 g (RoboBee) … 15 000 kg (Global Hawk)
speed	commercial: 50–150 km/h
energy	fuel · batteries (Li-Po) · solar cells
autonomy	**commercial battery drones: ≈ 30 min**
shape	single-rotor · multi-rotor · fixed wing · airship



RoboBee



fixed wing





Energy is the scarcest resource

As in sensor networks, but here **most of the energy goes into flying**, not into the radio

Why drones?



Search and rescue



Medical delivery



Fire monitoring



Infrastructure inspection



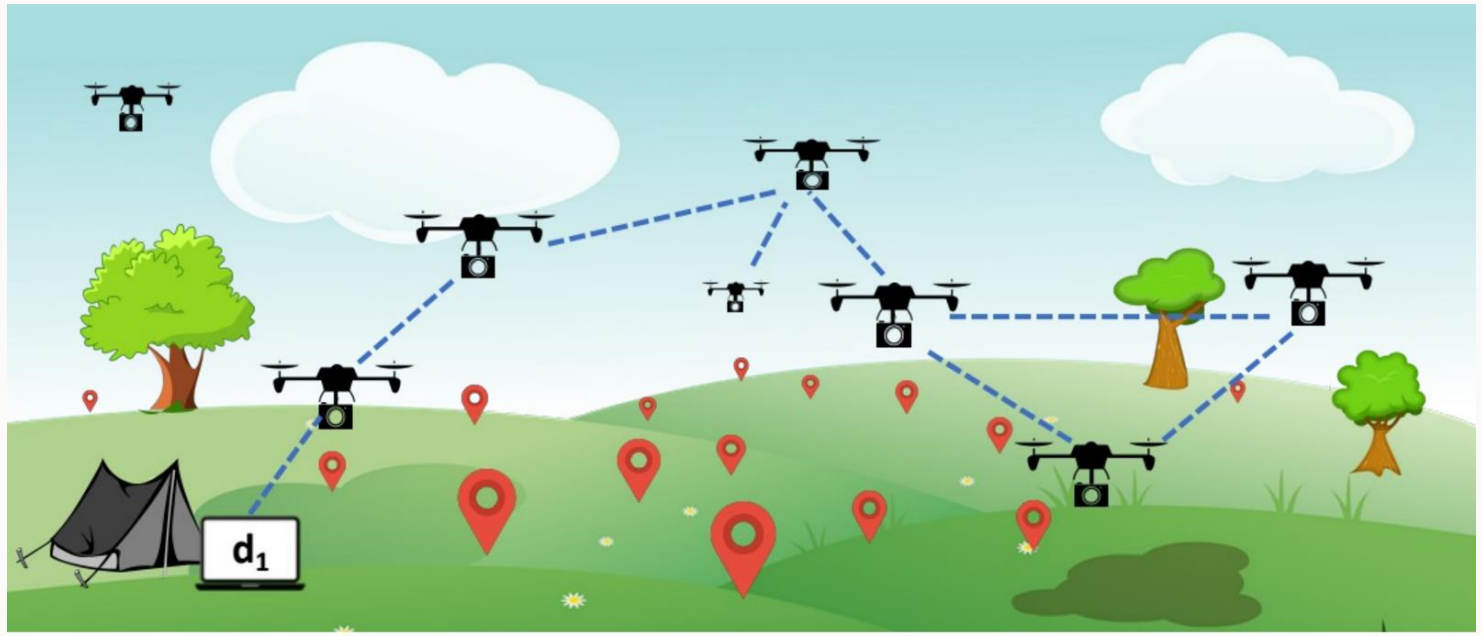
Precision agriculture



Emergency supplies

Common pattern: drones collect data over large or inaccessible areas, and the data must reach a ground station

From one drone to a network of drones



- **DRONET**, UAVNET or **FANET** (flying ad hoc network)
- Many drones fly over an area and cooperate on a mission
- Data flows to a **depot** (d_1): ground station and recharging point

The difficulty

Large areas and few drones: the network is often not connected

Dronets vs. wireless sensor networks

	WSN	Dronet
mobility	none or partial	**high, even 3D**
topology	random, stable	**mesh, changes in seconds**
infrastructure	absent (sink)	absent (depot)
energy	battery	**battery, very limited**
typical use	environmental monitoring	rescue, monitoring, surveillance

Moving costs more than transmitting

Flying: **≈ hundreds of W**

Wi-Fi radio: **≈ 1 W**

Orders of magnitude for a small multi-rotor

Challenges: medium-to-high mobility · fluid topology · node failures · energy constraints. Protocols designed for sensor networks rarely work as they are.

Four problems of networks of drones

1

Communication in emergencies

No infrastructure: drones as a temporary network

→ **DANGER**

2

Mission planning

Assign tasks and trajectories to drones that must recharge

→ **Trajectory selection**

3

Continuous data delivery

Multi-hop routing while the drones keep moving

→ **Routing protocols**

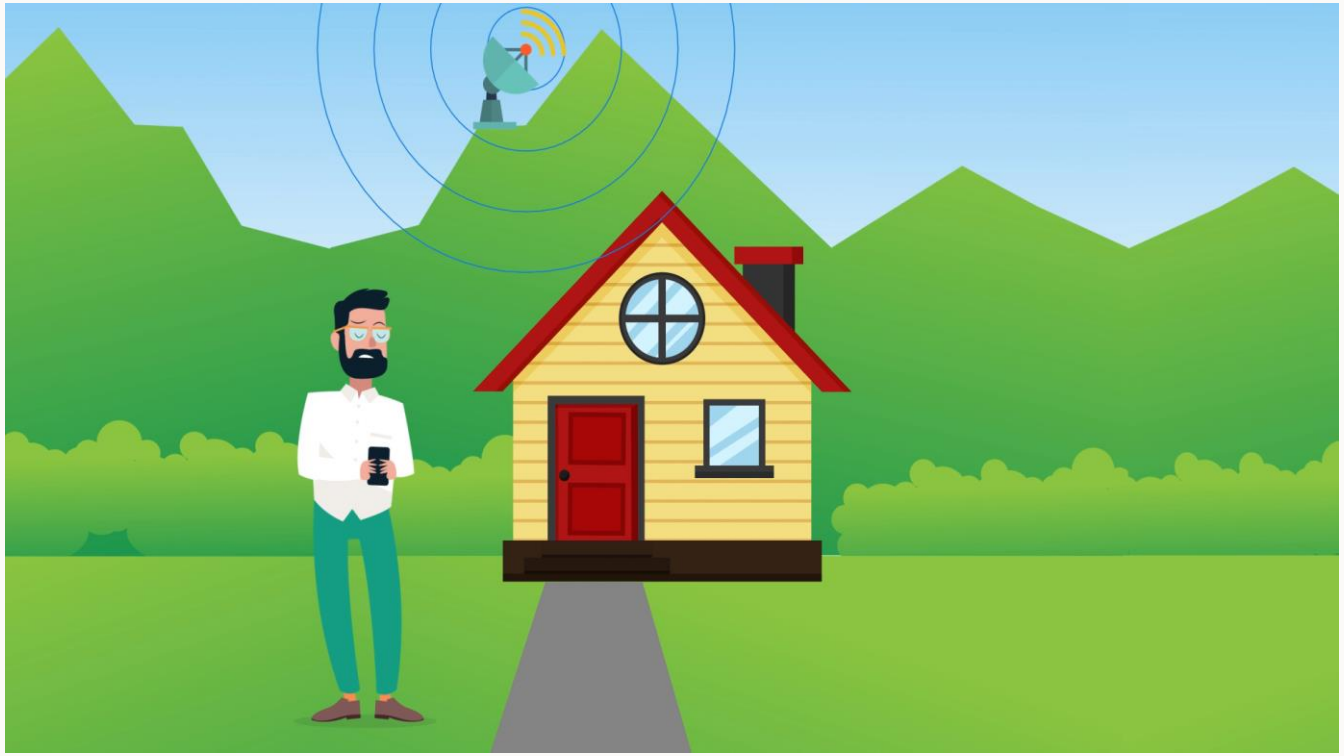
4

Periodic data offloading

Large areas: the drones are not always connected

→ **Stop & Route**

DANGER: drones as a temporary network



- Drones act as **flying access points and relays**
- **Goal:** easy to use communication
- Wi-Fi: any smartphone works
- no app to install, low cost

A. Coletta, G. Maselli, M. Piva, D. Silvestri, **ACM MobiHoc 2020**

DANGER: system components

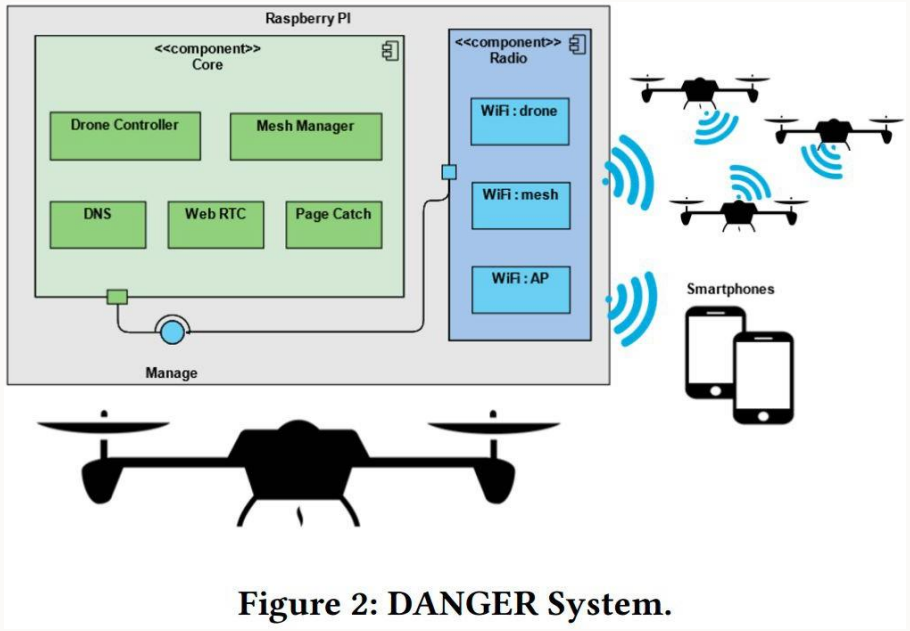
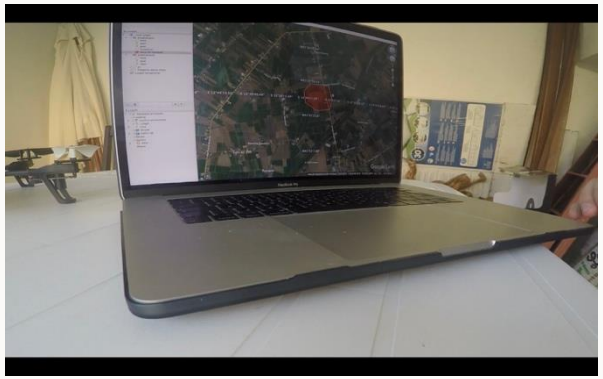


Figure 2: DANGER System.



Figure 3: DANGER Web Interface.



On the drone

Raspberry Pi: one Wi-Fi as access point, one for the mesh between drones

On the phone

Web interface, no app

At the base

Messages and positions on a map

DANGER in the field



PROBLEM 2 · MISSION PLANNING

Trajectory selection

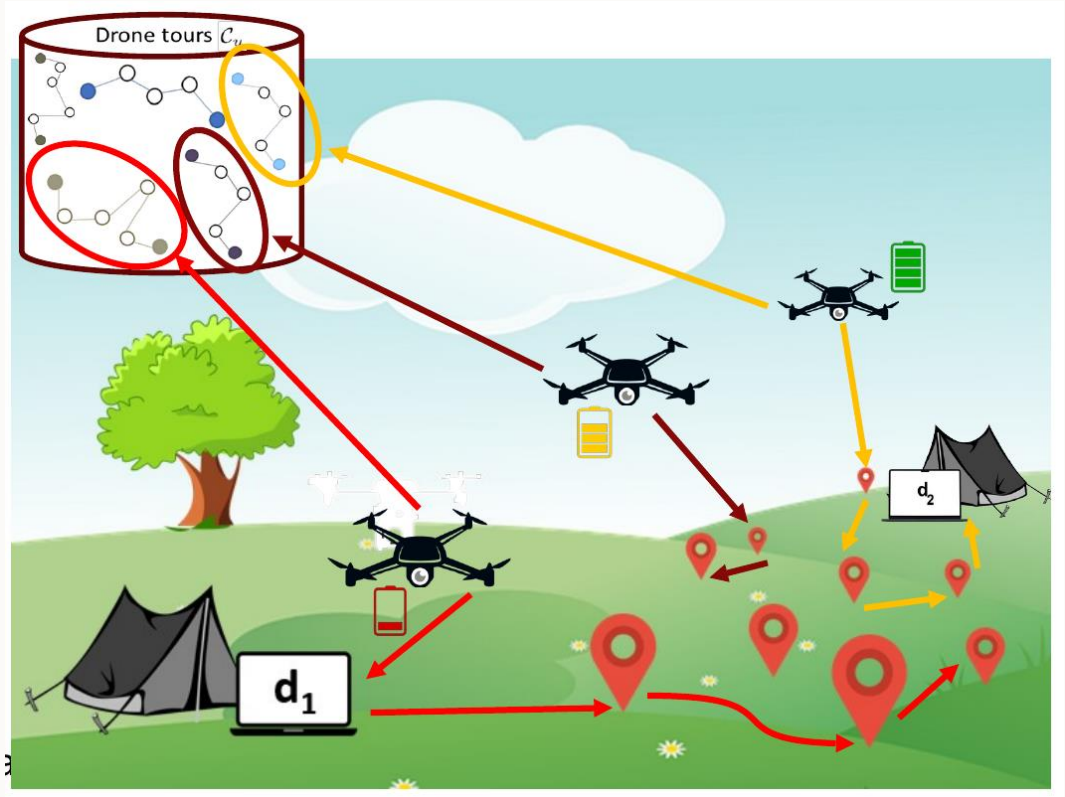
Problem

Several depots, a squad of heterogeneous drones (energy, capabilities, equipment)

Input: a set of trajectories and a required hovering time on each target

Goal

A multi-trip mission with recharging and offloading, not one trip at a time



WHERE WE ARE

Next: radios and MAC

Part 1 · Drones

- ✓ Drones and applications
- ✓ Dronets: four problems

Part 2 · Communication

- → **Radios and MAC**
- ○ Routing: families, B.A.T.M.A.N.
- ○ Geographic routing

Part 3 · Moving to connect

- ○ Stop & Route
- ○ From protocols to decisions

Radios on board: two channels

radio	range	data rate	channel access	role
Wi-Fi (802.11)	100s of m (line of sight)	up to 100s of Mbit/s	CSMA/CA	**primary: data**
LoRa (LPWAN)	km	0.3–50 kbit/s	ALOHA-like, duty cycle	**control channel**
cellular	wide area	high	scheduled by the base station	often not available

A common design

Wi-Fi for the data, LoRa for control messages (positions, commands). In Stop & Route simulations: **Wi-Fi 140 m, control channel 6 km**

Back to earlier lectures

CSMA/CA (the course project) and **ALOHA** (RFID lecture): the same MAC families, in the air

How long does a link between two drones last?

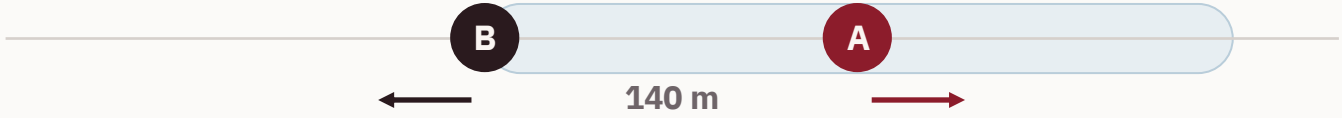
t = 0 s link starts: B enters the range of A



t = 8.75 s they cross



t = 17.5 s link breaks: B leaves the range of A



blue band: range of A (140 m on each side) · arrows: 8 m/s each

1. **Relative speed**
   v = 8 + 8 = 16 m/s

2. **Approach: until they cross**
   from 140 m to 0 m: 140 / 16 = 8.75 s

3. **Separation: link lifetime**
   from 0 m to 140 m: 8.75 s → T = 17.5 s

Head-on is the best case

Lateral distance d: $T = 2\sqrt{(r^2 - d^2)}/v$

d = 100 m → **≈ 12 s**

What changes for the MAC when nodes fly

Short-lived links

Relative speed 2 × 8 m/s, range 140 m:

link lifetime ≤ 2·140 / 16 = **17.5 s**

Neighbour tables age fast

More contention

Line of sight in the air: longer range, more nodes compete for the channel

Hidden terminals change continuously

Energy and overhead

Control messages cost bandwidth

But flying dominates the energy budget: avoid useless movements

FOR THE MAC

Same toolbox as before (CSMA/CA, ALOHA, duty cycling), but its parameters must **adapt to a topology that changes in seconds**. A natural place for learning.

PROBLEM 3 · CONTINUOUS DATA DELIVERY

Routing while the drones move

- Video and thermal streams need **wideband, short-range** radios (Wi-Fi)
- Wide areas → **multi-hop paths**: drones relay data for other drones
- Which routing protocol?

Scalability	**Energy conservation**
Low delay	**Low overhead**
Loop freedom	**High delivery ratio**

Goals in conflict

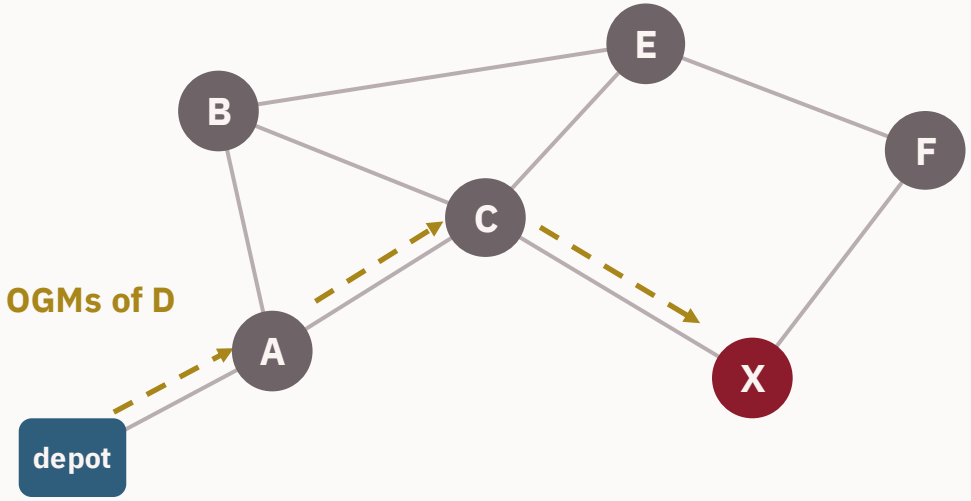
Fresh routes need control messages (overhead); fewer messages mean stale routes when the drones move

Four families of routing protocols

family	how	examples	for dronets
proactive	tables to all destinations, periodic updates	OLSR, DSDV, B.A.T.M.A.N.	overhead; slow to react to topology changes
reactive	route discovered on demand	AODV, DSR	discovery latency; routes break when drones move
hybrid	proactive nearby, reactive far away	ZRP, TORA	scales, but hard to implement
geographic	next hop chosen from positions (GPS)	greedy, GDSTR-3D	**no tables, no discovery: local decisions**

In this lecture: B.A.T.M.A.N. (proactive, driven by link quality) and geographic routing (greedy, store-carry-forward, prediction)

B.A.T.M.A.N.: routing driven by link quality



routing table of X	best next hop	TQ
depot	C	**72%**
B	C	81%
E	F	88%

Proactive, distance-vector for mobile ad hoc and mesh networks

Decentralised no node knows the whole topology

Next-hop based for each destination, only the best neighbour

Link-quality driven measured by counting OGM messages

Self-healing adapts when links appear and disappear

How B.A.T.M.A.N. works: originator messages

1. **Announce**
   Every node periodically broadcasts an OGM

2. **Flood**
   Neighbours rebroadcast OGMs: they reach the whole mesh

3. **Count**
   For each originator, count the OGMs arriving through each neighbour

4. **Choose**
   Best neighbour = next hop towards that originator

OGM field	purpose
originator address	who generated the OGM
sequence number	freshness, no loops
TTL	limits the hops
link quality (TQ)	quality towards the originator
hop count	hops travelled so far

The flaw of counting: asymmetric links

B.A.T.M.A.N. III: link quality = OGMs received



A counts B's OGMs
metric: 5% to B

B counts A's OGMs
metric: 100% to A

What happens

A → B: **100%** of the OGMs arrive

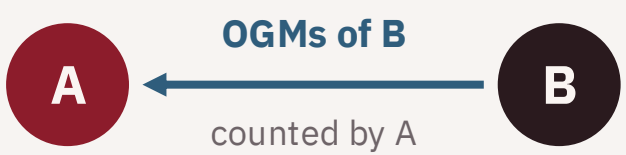
B → A: **5%** arrive

B counts only what it receives: it believes the link to A is perfect

The fix

Measure **both directions**: receive quality, echo quality, and their ratio, the transmit quality

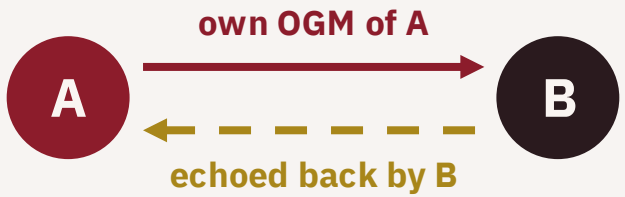
Receive, echo and transmit quality



Receive quality

RQ = # OGMs of B received by A

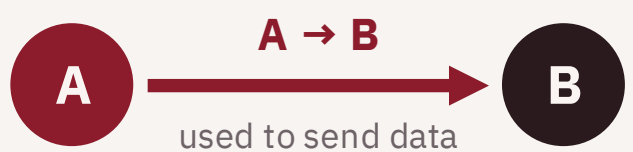
link B → A



Echo quality

EQ = # own OGMs echoed back by B

round trip A → B → A



Transmit quality

TQ = EQ / RQ

link A → B: the one used to send

COUNTED IN A SLIDING WINDOW

Only the last OGMs count (e.g. 64): the estimate forgets the past, like **sliding-window UCB** in RL lecture 3

Local TQ: the quality of one link

A counts over the last 64 OGMs	value
B's OGMs received by A	RQ = 48 packets
own OGMs echoed by B	EQ = 36 packets
transmit quality towards B	**TQ = EQ / RQ = 36 / 48 = 0.75**

Why the ratio works

EQ / 64 ≈ P(A→B) · P(B→A) an echo goes and comes back

RQ / 64 ≈ P(B→A) only the way back

EQ / RQ ≈ P(A→B) the direction A uses to send data

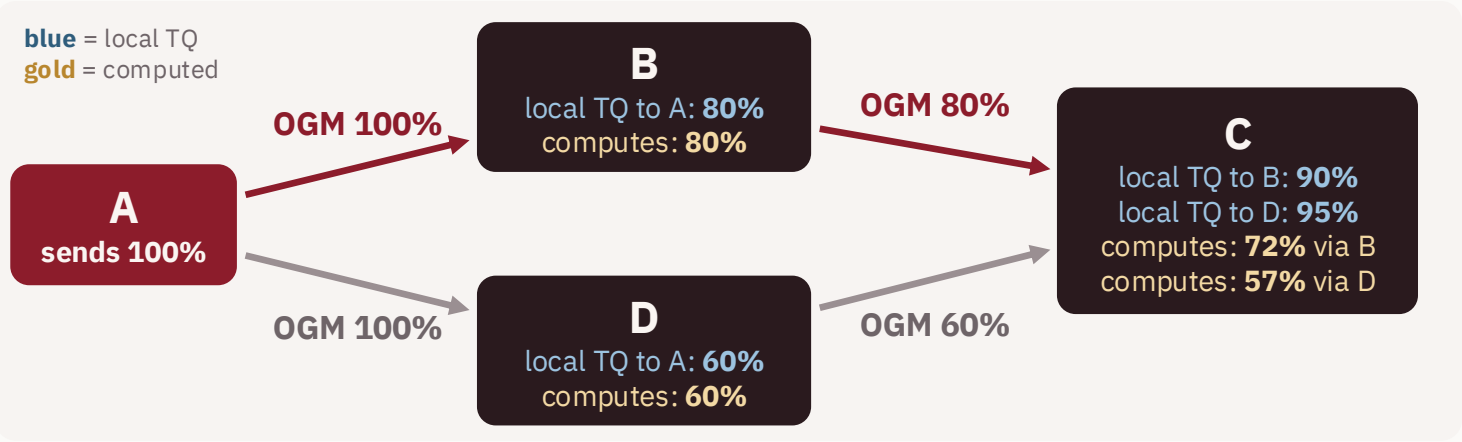
What 0.75 means

Of the packets A sends to B, about 75% arrive. The window size cancels out: RQ and EQ are packet counts

Local, link by link

A measures only the links to its own neighbours. It knows nothing about the links further away

Path TQ: carried hop by hop in the OGMs



node	receives (OGM)	× local TQ	= computes	then
B	100% from A	**80% (to A)**	**80%**	transmits 80%
D	100% from A	**60% (to A)**	**60%**	transmits 60%
C	80% from B	**90% (to B)**	**72%**	**best: next hop B**
C	60% from D	**95% (to D)**	**57%**	stored, not used

Propagation

$TQ = TQ_{received} \times TQ_{local}$

computed on reception, then transmitted in the rebroadcast OGM

Choosing the next hop

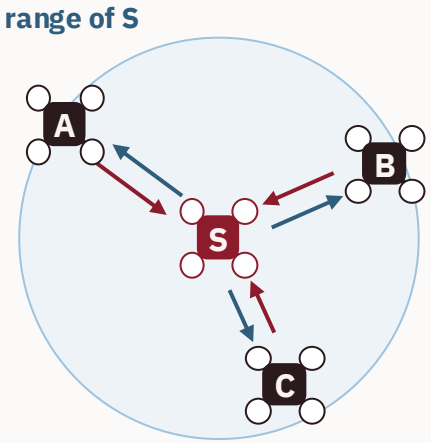
the neighbour whose OGM gives the highest computed TQ

WHAT C KNOWS

the end-to-end TQ per neighbour, not the single links: distance-vector

Geographic routing

- Each drone knows its **position** (GPS) and those of its neighbours (hello messages)
- No route discovery, **no routing tables**
- Only **local information**: less overhead, bandwidth and energy





E: out of range, no hello

→ hello from a neighbour
→ hello from S
hello = ID + position (x, y)

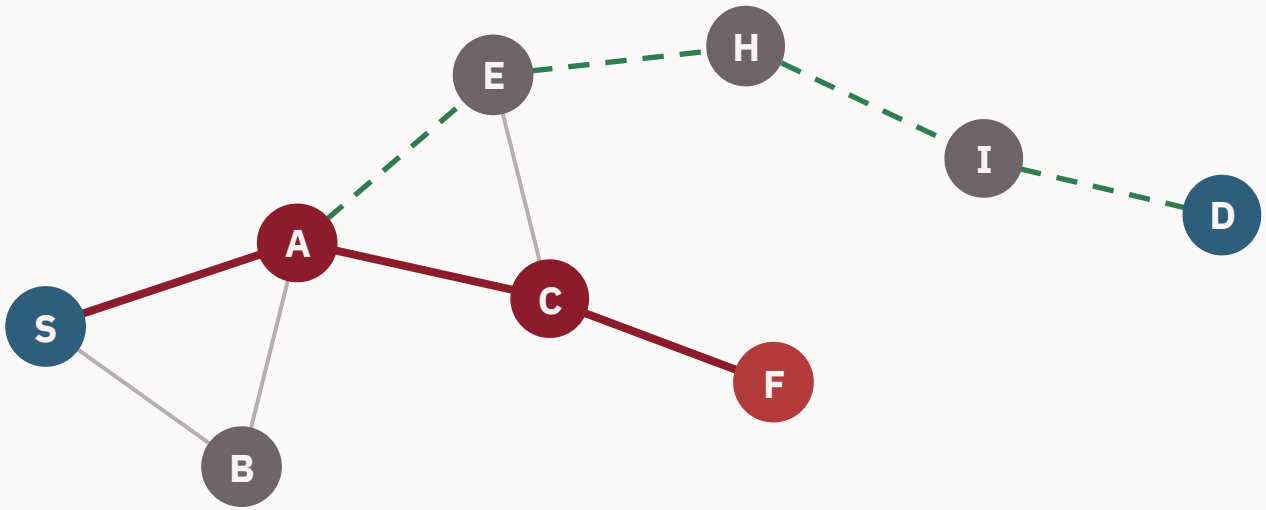
Greedy forwarding
closest neighbour to the destination

Store-carry-forward
no relay? carry the packet

Prediction
where will the neighbour be?

S's neighbour table	position (x, y)	last hello
A	(40, 120)	0.3 s ago
B	(190, 100)	0.8 s ago
C	(100, 10)	0.5 s ago

Greedy forwarding, step by step



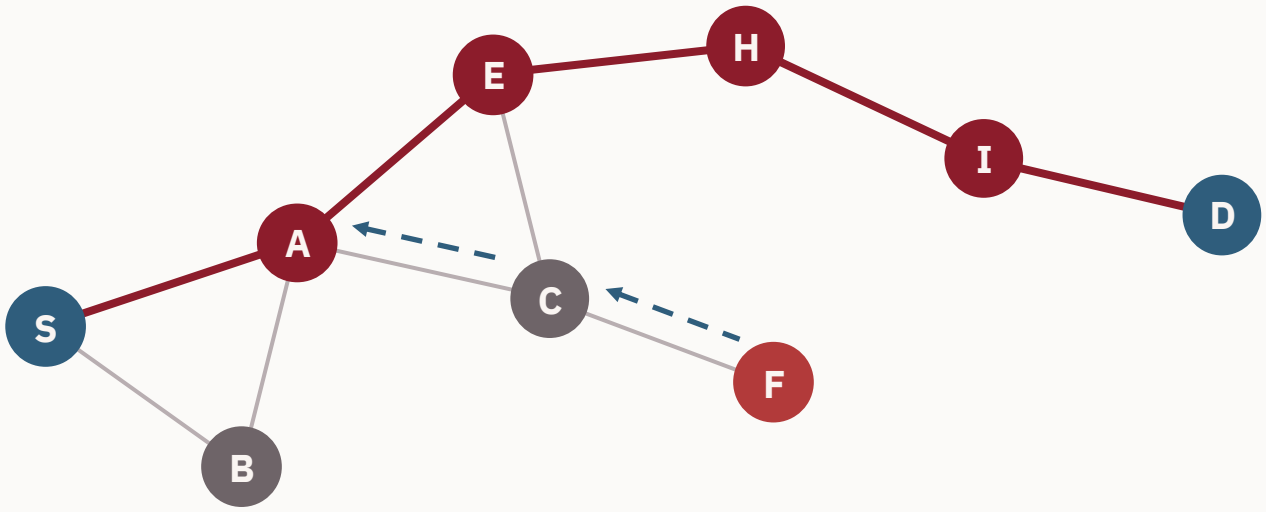
range 100 m · coordinates in m · **red: greedy path** · **green dashed: a path greedy misses**

at	neighbours: distance to D	choice
S (422)	A 330 · B 361	**A**
A (330)	C 242 · E 265 · B 361	**C**
C (242)	F 171 · E 265	**F**
F (171)	only C (242): none closer	**dead end**

Dead end (local optimum)

F is closer to D than all its neighbours, but D is out of range. A path exists (S-A-E-H-I-D), greedy misses it

1 · Backtracking: try another neighbour



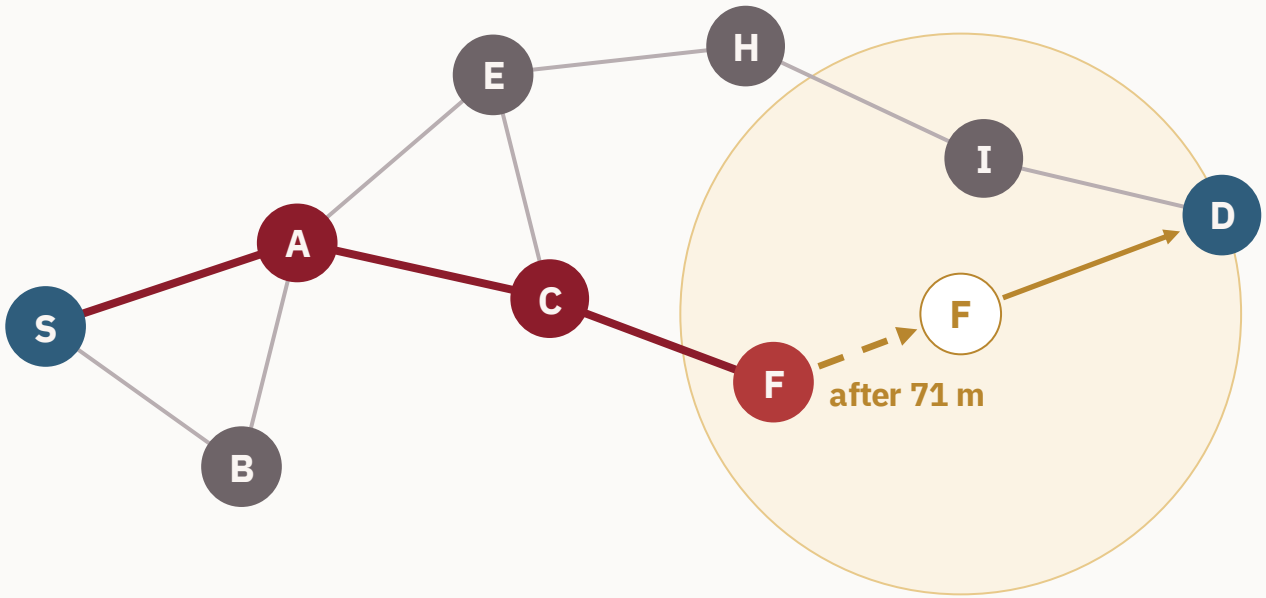
range 100 m · **blue: back** · **red: delivered path**

at	what happens
F (171)	no neighbour closer: back to C
C (242)	A, E farther than C: back to A
A (330)	next best neighbour: E (265)
E → H → I	greedy again: 180 → 87 → D

Cost

9 hops instead of 5 · extra transmissions · the packet must remember which neighbours were already tried

2 · Store-carry-forward: fly with the packet



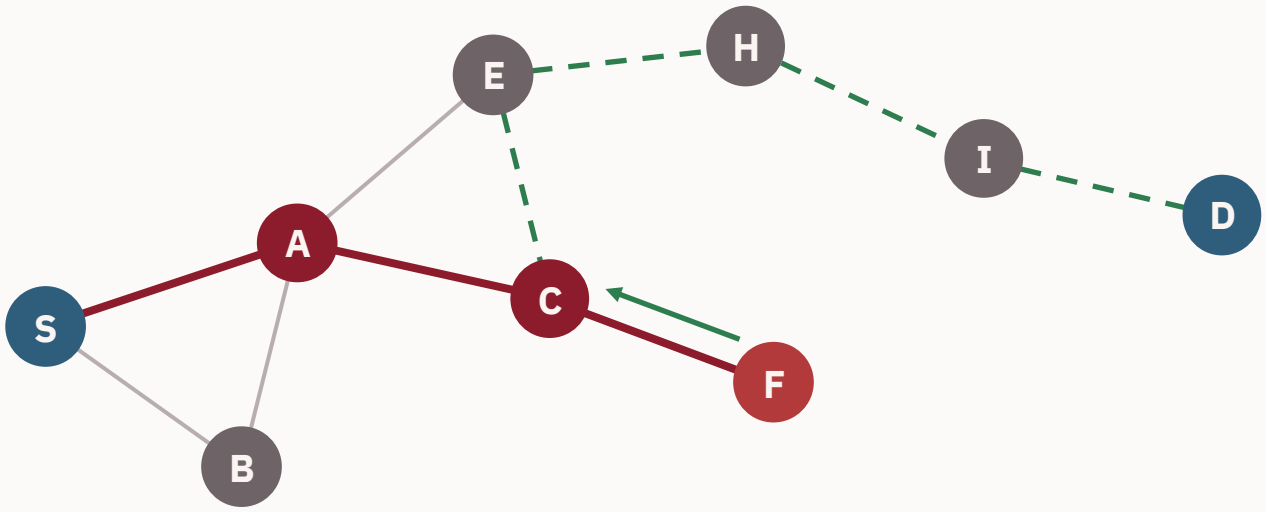
range 100 m · **red: greedy path** · **gold: F flies, then delivers**

step	what happens
1	F: no relay closer to D
2	F keeps the packet, flies towards D at 8 m/s
3	171 – 100 = 71 m ≈ 9 s: D in range
4	F delivers directly to D

Cost

no extra hops, but ≈ 9 s of delay · F leaves its mission position and spends energy · good for delay-tolerant data

3 · Face routing (GPSR): walk around the void



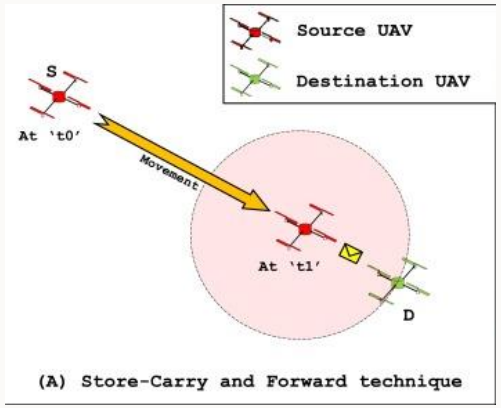
range 100 m · **red: greedy** · **green: around the void, then greedy**

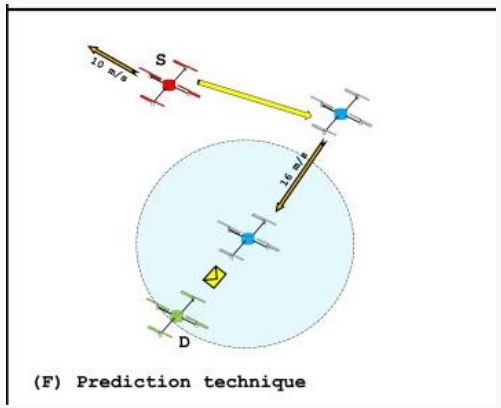
at	right-hand rule	vs 171 m
F	only edge → C	start
C	first edge ccw → E (not A)	—
E (265)	next edge → H	farther
H (180)	next edge → I	farther
I (87)	back to greedy → D	**closer**

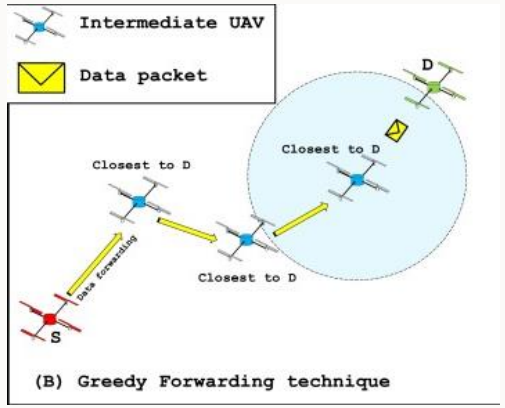
Cost

8 hops: takes C → E, which backtracking discards · delivery guaranteed only on a planar, static graph

Beyond greedy: when there is no relay







Store-carry-forward

No relay in range: carry the packet until meeting another drone or the destination. Delay traded for delivery

Prediction

Use position, direction and speed to choose the relay that will be closest to the destination

Combine

Greedy when possible, carry or predict at dead ends: no single rule works everywhere

WHERE WE ARE

Next: move the drones to connect

Part 1 · Drones

- ✓ Drones and applications
- ✓ Dronets: four problems

Part 2 · Communication

- ✓ Radios and MAC
- ✓ Routing: families, B.A.T.M.A.N.
- ✓ Geographic routing

Part 3 · Moving to connect

- → **Stop & Route**
- ○ From protocols to decisions

PROBLEM 4 · PERIODIC OFFLOADING

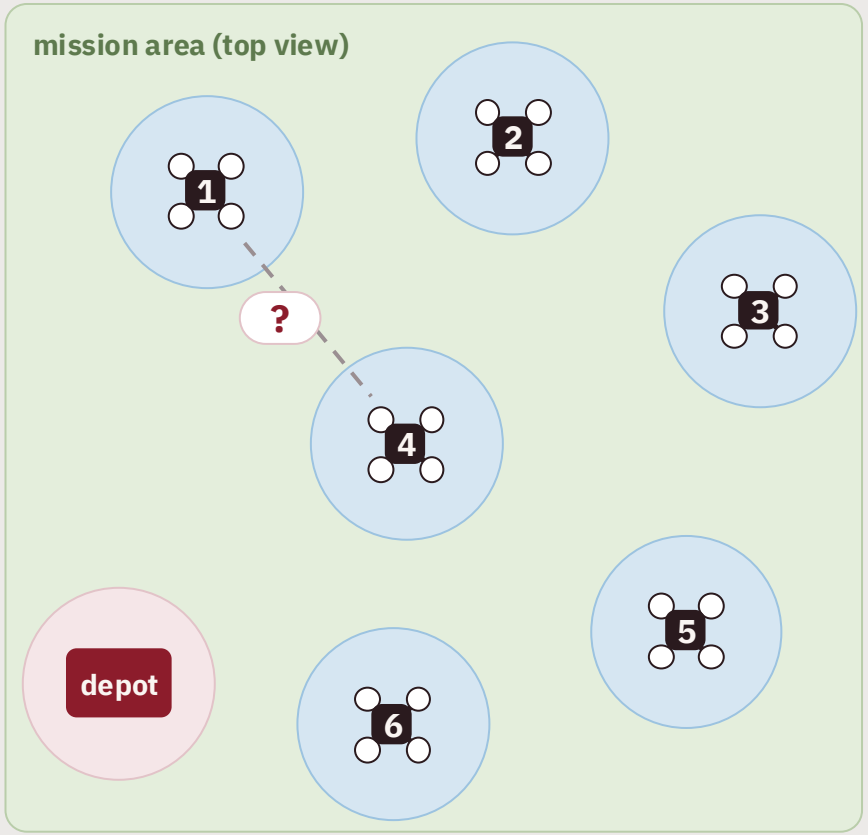
Try solving it yourself

A squad of drones senses a large area. There is a depot. Each drone has Wi-Fi (short range, high rate) and GPS. The drones are too far apart to be connected.

Periodically they must offload their data to the depot. Goals:

- minimise the distance travelled
- minimise the time to form a connected topology
- maximise the time spent on the mission

5 minutes, in pairs: where should each drone go? How does a drone know when to stop?



○ Wi-Fi range (140 m) · **no overlap: no path to the depot**

PROBLEM 4 · PERIODIC OFFLOADING

Stop & Route: stop, connect, offload, resume

Sense mission → **Stop** at fixed intervals → **Connect** move to a connected formation → **Offload** multi-hop to the depot

...and back to sensing: the cycle repeats until the batteries run out

Centralised (GAA, TCA)

A long-range control channel (LoRa): the depot collects the positions and computes the formation

Distributed (DGA)

Only short-range Wi-Fi: each drone decides from its neighbourhood

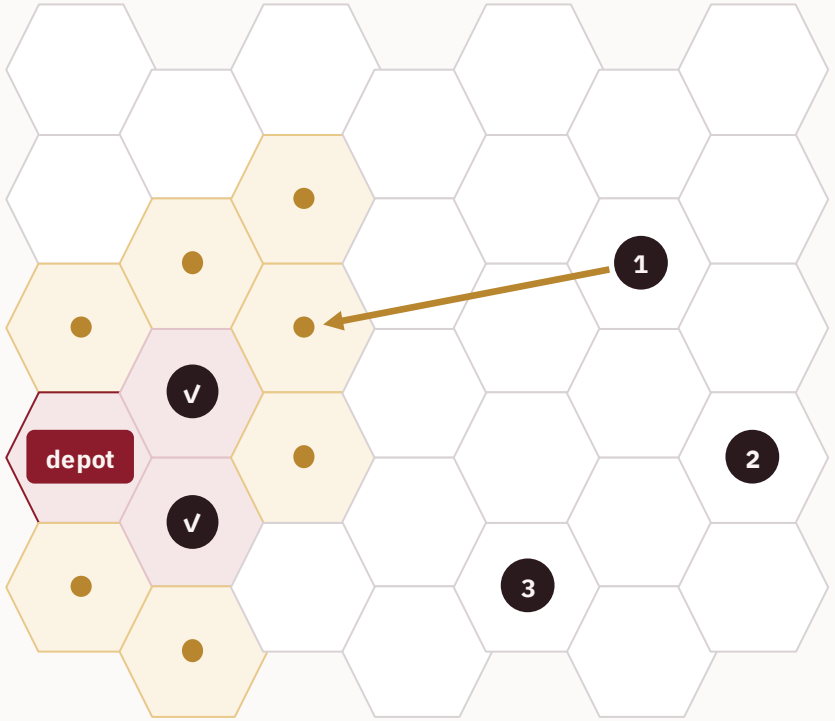
OPTIMAL FORMATION: MAXIMISE THE WORST RESIDUAL ENERGY

Choose the connected formation in which the weakest drone keeps the most battery

battery left = battery now – energy to fly to the new position – energy spent hovering while waiting

N. Bartolini, A. Coletta, F. Giorgi, G. Maselli, M. Prata, D. Silvestri · IEEE WONS 2023; Computer Communications 212, 2023

Optimal is too slow: the GAA heuristic



pink: connected region · **gold: candidate rendezvous points**

Why not the optimum?

an MILP: 18.6 s with 8 drones, 64 s with 25. Meanwhile every drone hovers and burns energy

GAA · Greedy Assignment Algorithm

1 Hexagonal tiles; rendezvous point at each centre

2 Connected region: starts from the depot

3 Candidates: free tiles next to the connected region

4 Assign the drone–tile pair that lowers the worst residual energy least

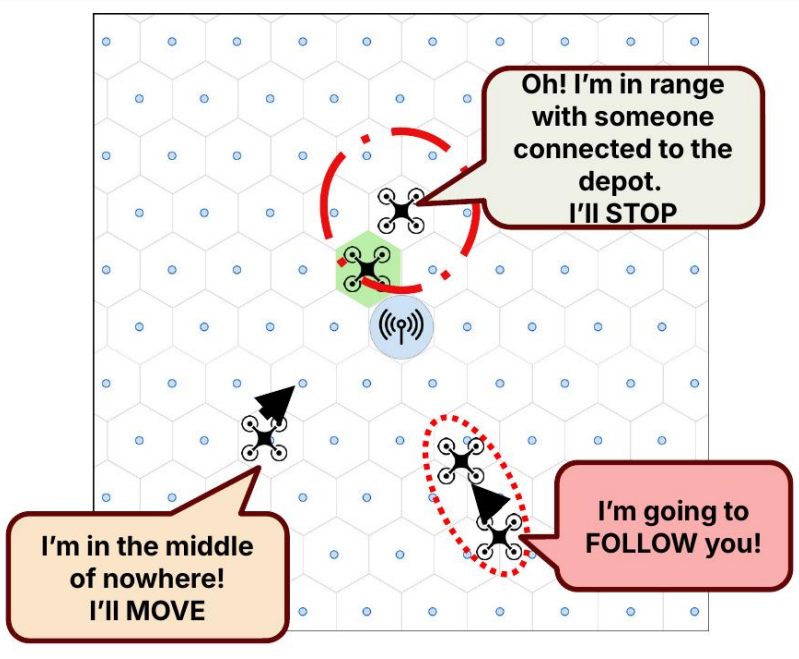
5 Repeat until every drone is connected

Polynomial: $O(m^5)$

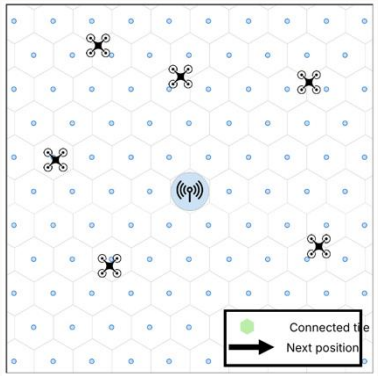
DGA: three local rules

- Area discretised: hexagonal tiles, rendezvous points at the centres
- Drones loosely synchronised: they all stop together
- Depot **heartbeats** (rebroadcast by connected drones) and drone **hellos** with positions

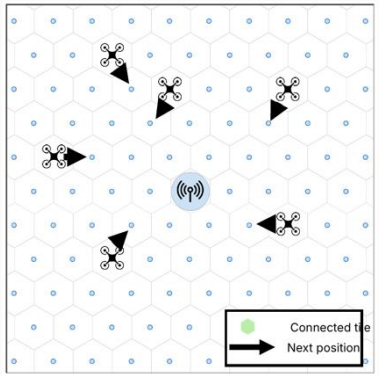
In range of the depot or of a connected drone	**STOP**
In range of a non-connected drone closer to the depot	**FOLLOW the leader**
Isolated	**MOVE towards the depot**



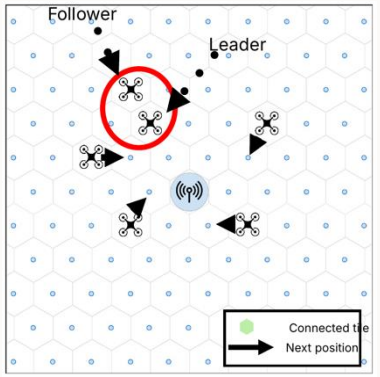
DGA, step by step



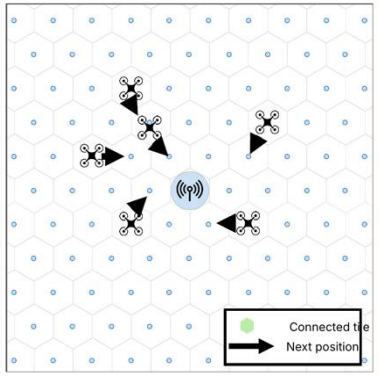
1 · Spread out



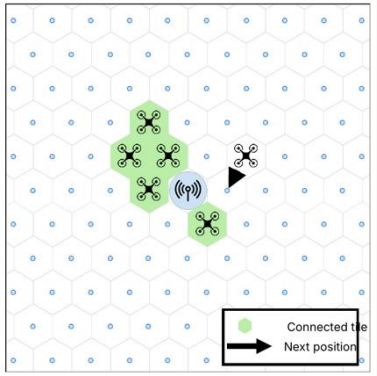
2 · Move closer



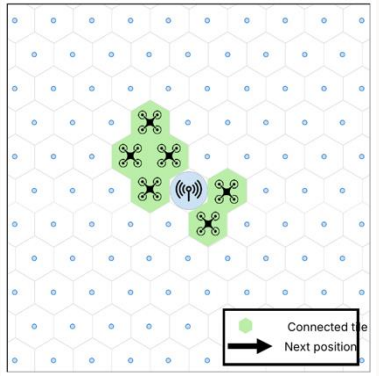
3 · Follow a leader



4 · First connections

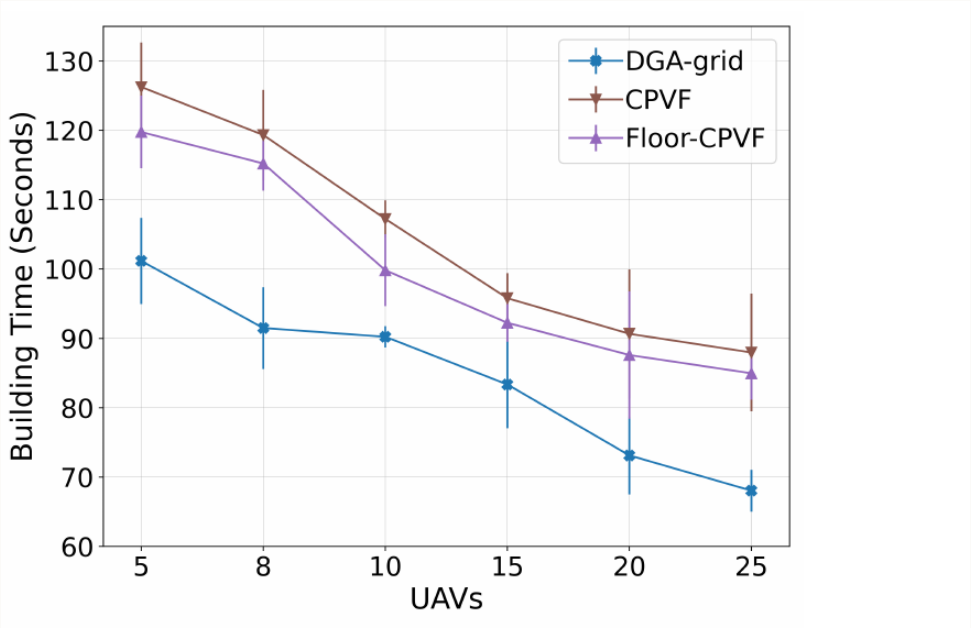


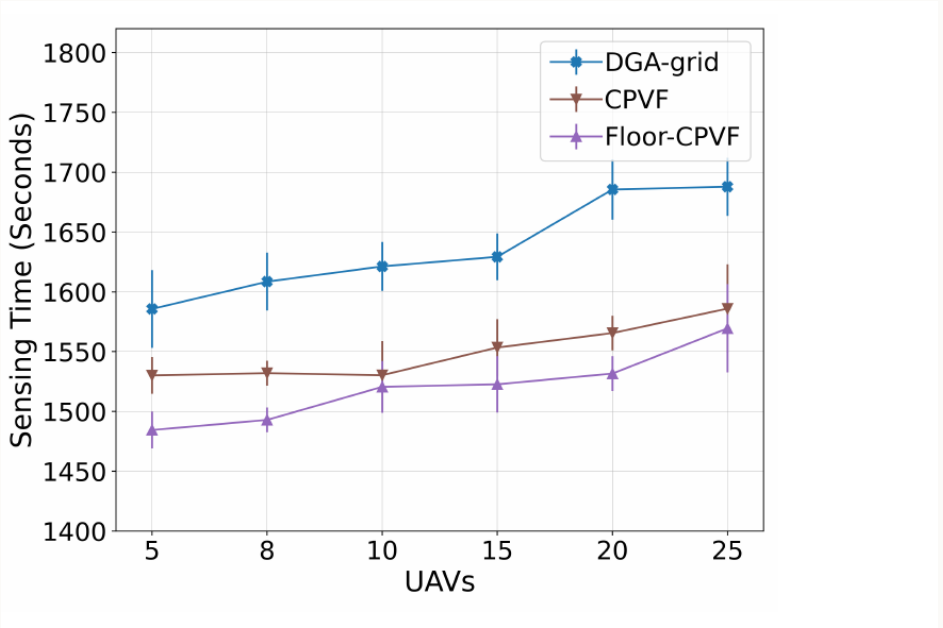
5 · Region grows



6 · All connected

Faster formations, more time for the mission





Building time

DGA-grid ≈ 100 s vs. > 120 s with 5 drones; ≈ 70 s vs. 85–90 s with 25

Sensing time

The time saved goes to the mission: ≈ +7% vs. CPVF

FROM PROTOCOLS TO DECISIONS

Every drone takes decisions

MDP concept	in a dronet
state	position, neighbours, queue, battery
action	forward to neighbour j · keep and carry · move towards the depot · keep sensing
reward	**packet delivered, delay, energy: arrives after several hops**
policy	greedy, store-carry-forward, DGA rules: written by hand

Bandit?

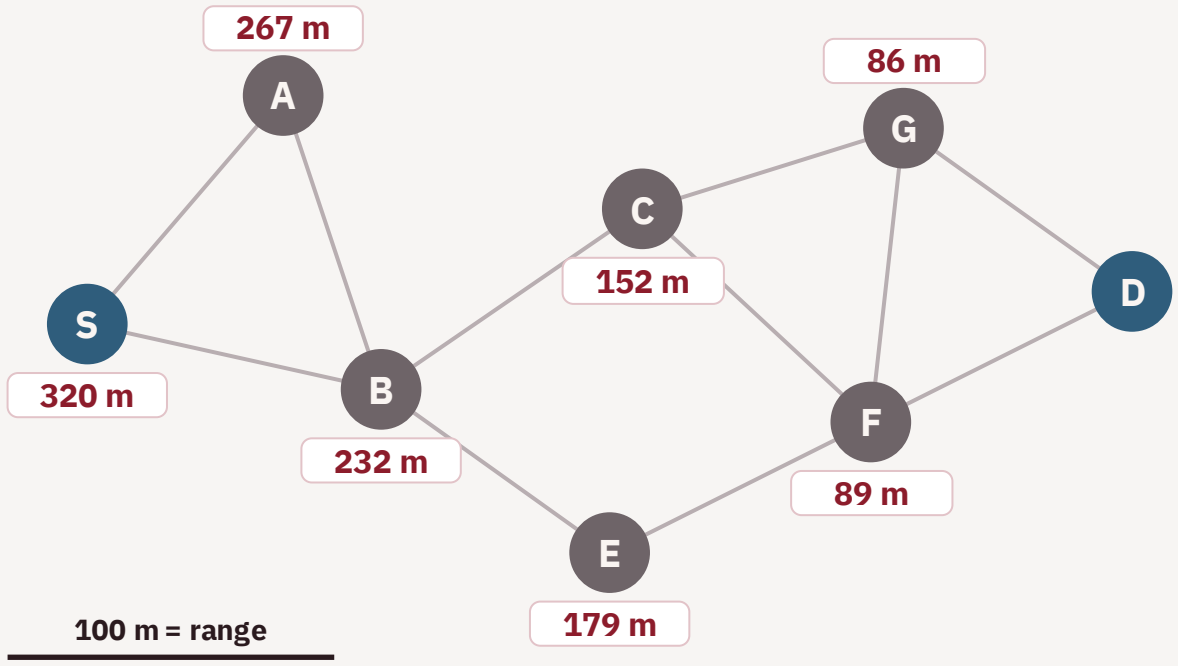
No: actions change the state, rewards are delayed

NEXT: MARKOV DECISION PROCESSES

Could a drone **learn** these policies? We need states, transitions and delayed rewards: **MDPs, then Q-learning**

EXAM PRACTICE · EXERCISE 1

Greedy geographic routing



Goal: greedy hop by hop, then see it fail: a new path (b), a dead end that only mobility solves (c)

Setting

S has a packet for D. A line joins two drones in range (100 m). The label of each drone is its distance from D

Rule: forward to the neighbour closest to D, only if it is closer to D than you

(a) Greedy path from S to D: justify each choice

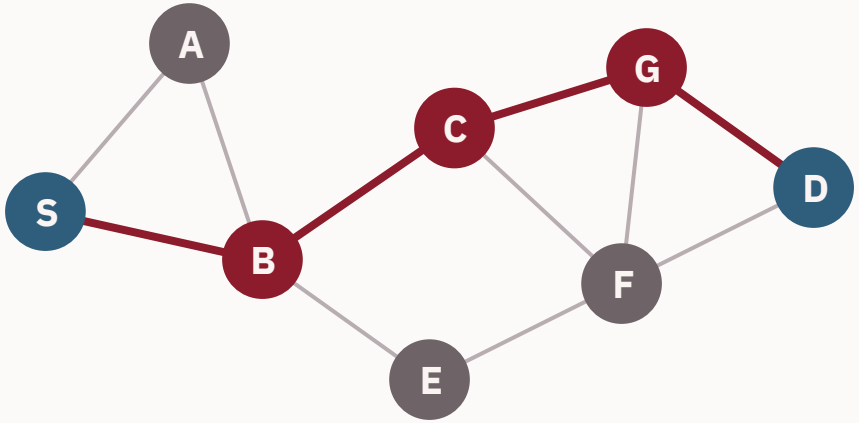
(b) G fails: new path?

(c) G and F fail: what happens? With store-carry-forward at 8 m/s, how long until delivery?

SOLUTION · EXERCISE 1

Distances to D decide every hop

at	d(·, D)	neighbours (≤ 100 m): distance to D	choice
S	320	A 267 · B 232	**B**
B	232	A 267 · C 152 · E 179	**C**
C	152	F 89 · G 86	**G**
G	86	D in range	**deliver**



(a) Path

S → B → C → G → D

(b) G fails

At C, F is at 89 m from D: S → B → C → F → D

(c) G and F fail

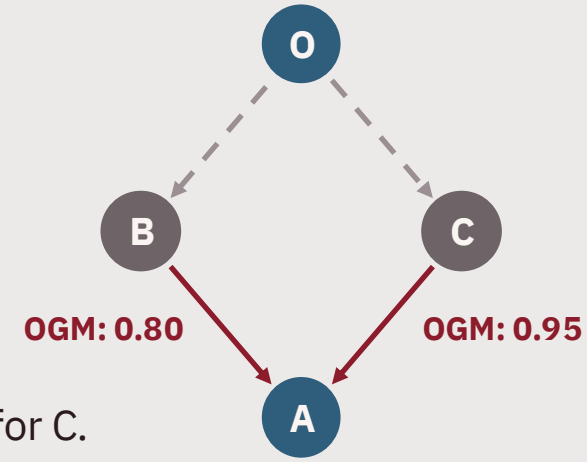
C is a dead end (152 m). Carry: (152 – 100) / 8 ≈ 6.5 s

EXAM PRACTICE · EXERCISE 2

B.A.T.M.A.N. link quality

Setting: node A wants a route towards the originator O. Its neighbours B and C both forward the OGMs of O. A counts the packets over the last 64 OGMs.

over the last 64 OGMs	neighbour B	neighbour C
OGMs of the neighbour received by A	60	64
own OGMs of A echoed by the neighbour	54	40
TQ towards O carried by the incoming OGM	0.80	0.95



(a) Give RQ and EQ (packet counts) and compute the local TQ = EQ / RQ for B and for C.

(b) Compute the TQ towards O through B and through C (TQ = TQ_received × TQ_local). Which next hop does A choose?

(c) Which neighbour would A choose looking only at the OGMs it receives? Why is that wrong?

SOLUTION · EXERCISE 2

C looks best, B is better

	neighbour B	neighbour C
RQ (packets)	60	**64 (all)**
EQ (packets)	54	40
local TQ = EQ / RQ	**54 / 60 = 0.9**	40 / 64 = 0.625
TQ towards O	**0.80 × 0.9 = 0.72**	0.95 × 0.625 ≈ 0.59

(b) Next hop: B

0.72 > 0.59, even though the OGMs through C carried 0.95

(c) Counting only: C

RQ = 64 of 64 looks perfect, but A → C loses over a third of the packets: the asymmetric-link mistake

EXAM PRACTICE · EXERCISE 3

How long does a link last?

Drones fly at 8 m/s; the Wi-Fi range is 140 m.

(a) Two drones fly in opposite directions and cross each other. For how long, at most, can they communicate?

(b) A drone is 600 m from the depot, with no other drone in between. How long does it take to deliver its data by carrying it (store-carry-forward)?

(c) What do (a) and (b) tell us about routing in a dronet? Why does Stop & Route stop the drones before offloading?

SOLUTION · EXERCISE 3

Links last seconds, carrying takes a minute

(a) Link lifetime

relative speed 8 + 8 = 16 m/s

in range over 2 × 140 = 280 m

280 / 16 = **17.5 s** at most

(b) Carry the data

fly until the depot is in range:

600 − 140 = 460 m

460 / 8 = **57.5 s**

(c) Lesson

Routes must be used within seconds; carrying costs delay, flying energy and sensing time

WHY STOP & ROUTE STOPS

In a connected formation the drones hover: **static topology, stable links**, and the data flows at Wi-Fi rate along multi-hop routes

CONCEPT CHECK

True or false?

T / F ?	**1.** Proactive routing reacts quickly to topology changes in a dronet.
T / F ?	**2.** Greedy geographic forwarding always finds a path, if one exists.
T / F ?	**3.** B.A.T.M.A.N. evaluates a link only by counting the OGMs it receives.
T / F ?	**4.** Stop & Route uses the controlled mobility of the drones to connect the network.

CONCEPT CHECK

True or false: answers

FALSE

Proactive routing reacts quickly to topology changes in a dronet.
Tables are refreshed by periodic messages: overhead, and slow reaction.

FALSE

Greedy geographic forwarding always finds a path, if one exists.
It can stop at a dead end even when another path exists.

FALSE

B.A.T.M.A.N. evaluates a link only by counting the OGMs it receives.
That fails on asymmetric links: TQ = EQ / RQ measures the sending direction.

TRUE

Stop & Route uses the controlled mobility of the drones to connect the network.
Drones move to a connected formation, then offload along multi-hop routes.

CONCLUSIONS

How do we move data across a network whose nodes fly?

Use the network

Links last seconds. Routing with local information (geographic) or link quality in a sliding window (B.A.T.M.A.N.); greedy can hit dead ends

Move the nodes

Carry the data, or move to a connected formation: Stop & Route saves building time and gives time back to the mission

ALL HAND-WRITTEN POLICIES

Each protocol decides from the drone's current situation: a **state**

NEXT

Markov decision processes: states, actions and delayed rewards. Then the drones learn the policies